

Mathematical Analysis and Performance Metrics of the Wave-Particle Transport Simulation (WTS) Rendering System

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January 9, 2026

Abstract

This document presents a mathematical derivation and performance analysis of the Wave-Particle Transport Simulation (WTS) rendering system. We analyze the system’s core hybrid architecture, which couples Signed Distance Function (SDF) ray marching with a grid-based, non-linear hyperbolic partial differential equation (PDE) solver. We demonstrate that the WTS “relaxation” kernel functions as a programmable screen-space wave propagator, effectively approximating global illumination phenomena such as subsurface scattering and bloom through continuous media mechanics rather than discrete Monte Carlo integration. We compare the computational complexity (O) and visual convergence properties against State-of-the-Art (SOTA) path tracing methods.

1 Introduction

Standard real-time ray tracing (RTX) relies on the light transport equation, typically solved via Monte Carlo integration. While physically accurate, this method suffers from high variance (noise) and requires significant temporal accumulation or AI denoising (e.g., DLSS, OptiX) to converge.

The WTS framework introduces a novel approach: **Axiomatic Physical Homeostasis (APH)**. Instead of treating light transport purely as ray optics, the system treats the screen buffer as a reactive medium governed by wave mechanics. The ray tracer acts as a “source term” injecting energy into a continuous simulation, which then propagates flux according to local stiffness constraints derived from the scene geometry.

2 Mathematical Formulation

2.1 The Ray Marching Source Term

The primary visibility determination is performed via Sphere Tracing on Signed Distance Fields (SDF). For a camera origin $\mathbf{o}$ and ray direction $\mathbf{d}$, the intersection distance t^* is found by iteratively stepping:

$$t_{i+1} = t_i + \min_k (SDF_k(\mathbf{o} + t_i \mathbf{d})) \quad (1)$$

This pass populates two critical buffers:

- $C_{in}(x, y)$: The raw color/energy injected into the system.
- $\beta(x, y)$: The *Stiffness Map*, representing the local medium’s resistance to flux.

2.2 The WTS Wave Equation

The core innovation lies in the `wts_relax_step` kernel. By analyzing the discrete update logic provided in the source code, we can derive the continuous partial differential equation it approximates.

The code calculates a “stress” tensor based on the Laplacian of the buffer Ψ :

$$\text{Stress}_{i,j} = \nabla^2 \Psi_{i,j} \approx \Psi_{i-1,j} + \Psi_{i+1,j} + \Psi_{i,j-1} + \Psi_{i,j+1} - 4\Psi_{i,j} \quad (2)$$

The force applied to the momentum field P is non-linear, governed by a hyperbolic tangent saturation function:

$$F = \alpha \tanh \left(\frac{\nabla^2 \Psi}{\beta(x, y)} \right) \quad (3)$$

where $\beta(x, y)$ is the local vacuum/object stiffness.

The time-integration follows a semi-implicit scheme with damping γ :

$$\frac{\partial P}{\partial t} = F - \gamma P \quad (4)$$

$$\frac{\partial \Psi}{\partial t} = P + S_{in} \quad (5)$$

Combining these, we obtain the **WTS Non-Linear Damped Wave Equation**:

$$\frac{\partial^2 \Psi}{\partial t^2} + \gamma \frac{\partial \Psi}{\partial t} = \alpha \tanh \left(\frac{\nabla^2 \Psi}{\beta(\mathbf{x})} \right) + \lambda S_{in}(\mathbf{x}, t) \quad (6)$$

2.2.1 Stability Analysis (APH)

Standard wave equations often diverge (explode) with high energy injection. The WTS framework enforces stability via the tanh saturation term.

- As $\nabla^2 \Psi \rightarrow 0$, $\tanh(x) \approx x$, behaving like a linear diffusion/wave equation.
- As $\nabla^2 \Psi \rightarrow \infty$, $\tanh(x) \rightarrow 1$, limiting the maximum force applied per frame.

This guarantees $\left| \frac{\partial^2 \Psi}{\partial t^2} \right| \leq \alpha$, ensuring the system remains bounded regardless of input intensity (lightning/explosions), fulfilling the ‘‘Homeostasis’’ axiom.

3 Performance Metrics vs. SOTA

We compare the WTS-RTX approach against standard hardware-accelerated Path Tracing (PT) and Screen Space Global Illumination (SSGI).

Metric	Path Tracing (SOTA)	SSGI / SSAO	WTS-RTX (Ours)
Complexity	$O(N_{rays} \cdot \log(N_{tris}))$	$O(N_{pixels} \cdot K_{kernel})$	$O(N_{pixels})$
Light Transport	Physically Accurate	Screen-Space Approx.	Wave-Based Propagation
Noise Characteristic	High Frequency (Grain)	Low Freq (Banding)	Smooth / Fluid-like
Convergence Time	Frames to Seconds	Instant (1 Frame)	Continuous (Time-Integrated)
Memory Bandwidth	High (BVH Traversal)	Medium (G-Buffer)	Low (Ping-Pong Buffer)

Table 1: Comparative Analysis of Rendering Architectures

3.1 Computational Complexity

- **Path Tracing:** Performance scales linearly with the number of light bounces. Achieving glow or subsurface scattering requires many rays, degrading performance significantly ($O(N_{bounces})$).
- **WTS-RTX:** The cost of the relaxation step is constant per pixel, regardless of scene complexity. The ‘‘bounces’’ of light are simulated by the wave propagation over time.

$$T_{WTS} \approx T_{PrimaryRay} + T_{Relaxation} \quad (7)$$

where $T_{Relaxation}$ is negligible compared to BVH traversal.

3.2 Memory Coherency

SOTA Ray Tracing suffers from incoherent memory access (rays scatter in random directions, thrashing the cache). WTS-RTX relies on neighbor-sampling (stencil operations), which is highly cache-coherent and optimized for GPU SIMD architectures (Warp/Wavefront execution).

4 Visual Artifact Analysis

The WTS system exhibits unique visual characteristics distinct from standard rendering:

1. **Interference Patterns:** Unlike Monte Carlo noise, WTS artifacts manifest as Moire patterns or standing waves. This is physically consistent with light behaving as a wave.

2. **Light Bleeding:** The variable stiffness β allows light to “tunnel” through objects or wrap around corners, naturally approximating diffraction and subsurface scattering without explicit coding.
3. **Temporal Inertia:** Changes in lighting (e.g., lightning) do not vanish instantly but decay over time due to the momentum term P . This provides free temporal anti-aliasing and motion blur.

5 Conclusion

The WTS-RTX system represents a paradigm shift from geometric optics (Ray Tracing) to wave optics (Field Simulation). By formulating the framebuffer as a continuous medium obeying the *Axiomatic Physical Homeostasis* protocols, the system achieves complex global illumination effects—such as bloom, color bleeding, and refraction—at a computational cost significantly lower than equivalent path-tracing estimators. While it sacrifices strict geometric accuracy for energetic plausibility, it offers a robust, stable, and highly performant alternative for real-time visual synthesis.